



PUSL3190 Computing Project

Interim Report

**Local Browser Isolation With Download File
Activity Monitoring**

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Chapter 01: Introduction

The modern threat landscape reveals browsers as the primary attack vector for cyber threats, with

- *Browser-based phishing* attacks increasing 198% year-over-year
- Reason for 12% of all cyberattacks globally (Menlo Security, 2024; Keepnet, 2025).

Traditional security approaches fail to provide protection against the *browser-based attack* or *APT*. Because the antivirus software is *signature based* and *process on the pre-collected data or a malware*.

- Antivirus software *misses 30% of browser-based evasive threats*.

while *Remote Browser Isolation (RBI)* solutions cost enterprises \$120,000-\$600,000 annually for 1,000 users, making it expensive for most organizations (Venn, 2025).

Container-based local isolation using *Docker* or similar technologies *shares the host operating system kernel*, creating *vulnerabilities to kernel-level exploits where malware can escape containment* and compromise the host system.

My project develops a *Local browser isolation system* that provides *hardware-level security without cloud infrastructure costs*. The system isolates browsing activity in *lightweight virtualization* environments while implementing comprehensive *file monitoring to detect and block malicious activities*. By eliminating cloud cost and maintaining native-like performance, the solution addresses the cost and usability limitations that prevent widespread browser isolation adoption.

Problem Definition

Current browser security solutions create a *critical trade-off between security effectiveness and operational practicality*. *RBI provides strong isolation*, but *latency is high* when it comes to complex tasks, *high infrastructure costs* thousands of dollars monthly per deployment, and *compatibility failures* with approximately 15-20% web applications that use *dynamic JavaScript and complex rendering* (StrongDM, 2025).

Application functionality and *productivity losses estimated at 10-15% in high-bandwidth tasks* (Menlo Security, n.d.).

Current browsers using traditional containers share the host operating system kernel (Google Chrome Site Isolation / Firefox Sandbox), where a single container can potentially affect all others on the system and vulnerability to kernel-level exploits. (HuggingFace, n.d.).

The *Docker container* approach, while lightweight, has documented cases of *container escape vulnerabilities* that allow malware to *breach isolation and access the host system directly* (Medium, 2024).

Current file monitoring solutions often fail to detect sophisticated malware that employs anti-sandbox detection techniques, with estimates suggesting 560,000 new pieces of malware are detected daily (Cybersecurity Ventures, 2024).

We aim to develop a local browser isolation system using Firecracker-microVMs. Because it provides hardware-level isolation, running a lightweight virtual machine with its own kernel, offering stronger security and protection.

The system will implement real-time file monitoring with behavioral analysis to detect malicious activities, maintaining detailed logs of all browser and file system interactions. The solution targets minimal resource overhead (under 5MB per isolated session) and native browsing performance comparable to standard browsers, eliminating the latency and cost issues inherent in cloud-based RBI solutions.

Project Objectives

- Design and implement a local browser isolation system using *lightweight virtualization* that *achieves complete process and kernel-level separation* from the host operating system.
- *Develop a dual-layer file monitoring system* consisting of *real-time activity tracking* and *post-download analysis* with behavioral threat detection.
- Create an intuitive user interface that enables isolated browsing sessions with a single click, *hiding underlying complexity from end users*.
- *Implement behavioral analysis engine* using *rule-based detection* for ransomware, information stealers, and malicious scripts.
- Achieve browser rendering performance within native browser speeds.

Chapter 02: System Analysis

Existing System

Currently we can divide browser security implementations into three categories:

1. **Remote Browser Isolation (RBI):** Cloud-based systems where browsing occurs on remote servers with only safe visual representations transmitted to user devices. Solutions from vendors like Menlo Security, Palo Alto Networks, and Zscaler provide strong isolation but introduce *significant latency (2-5 seconds per page load)*, *high bandwidth consumption (5-10 Mbps per session)*, and *substantial costs (\$120,000-\$600,000 annually for 1,000 users)*.
2. **Container-based Local Isolation:** Solutions using Docker or similar technologies (Podman) to isolate browsers in containers on user devices. While *lightweight* and *fast*, these *share the host operating system kernel*, creating *vulnerabilities to kernel-level exploits*. Containers escape vulnerabilities documented in CVE databases proving that *malware can breach container boundaries and compromise host systems*.
3. **Traditional Endpoint Security:** Antivirus software and endpoint detection systems that monitor browser activity without isolation. These reactive approaches *miss 30% of browser-based evasive threats* and provide *no containment if malware executes successfully*. Cannot prevent zero-day exploits or sophisticated phishing attacks.

Drawbacks of the Existing System

Remote Browser Isolation Limitations:

- High latency degrading user experience.
- Significant bandwidth requirements.
- High costs.
- Deployment complexity requires specialized staff.
- Compatibility issues with modern web applications.
- Poor performance for multimedia content and high-DPI displays.
- Cannot prevent social engineering attacks targeting user behavior.

Container-based Isolation Limitations:

- Shared kernel vulnerability allowing container escape attacks.
- No hardware-level isolation.
- Limited visibility into malicious behaviors within containers.
- Kernel vulnerabilities affect all containers simultaneously.
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Traditional Endpoint Security Limitations:

- No isolation or containment of threats.
- Signature-based detection misses zero-day exploits.
- Behavioral analysis is insufficient for sophisticated evasion techniques.
- Reactive rather than preventive security model.
- Cannot protect against drive-by downloads and browser exploits.

Chapter 03: Requirements Specification

Functional Requirements

1. Browser Isolation Management

- System shall launch isolated browser instances in separate virtualization environments.
- Complete filesystem and process isolation from host operating system.
- Manage browser lifecycle (creation, execution, termination).
- Support multiple simultaneous isolated browsing sessions.

2. File Monitoring and Analysis

- Monitor all file operations within isolated environment in real-time.
- System track file creation, modification, deletion, and execution attempts.
- Analyze downloaded files before allowing transfer to host system.
- System maintain detailed audit logs of all file system activities.

3. Threat Detection

- Detect threat indicators
 - Rapid encryption, mass file modifications.
 - Identify information stealer behaviors (credential access, clipboard monitoring)
 - Recognize privilege escalation attempts and unauthorized system modifications
- Block malicious file operations and alert users to detected threats.

4. User Interface

- Provide simple launcher application for initiating isolated browsing.
- Display browser directly without exposing underlying virtualization.
- Enable file downloads through controlled transfer mechanism.

5. Logging and Reporting

- Maintain comprehensive logs of browsing activities and security events
- Generate threat reports with forensic details for security analysis
- Export logs in standard formats for integration with security tools.

Non-Functional Requirements

6. Performance

- Browser opening time: less than 10 seconds from launch to usable browser.
- Page rendering with native browser performance.
- Memory overhead within 100MB per isolated session.
- Less CPU overhead for monitoring and isolation processes.

7. Security

- Complete kernel-level isolation preventing malware escape to host.
- File monitoring with known malware families.
- Secure communication between isolated environment and host.

8. Usability

- Familiar browser experience without specialized training.
- Minimal user effect for file downloads and getting to the host system.

9. Compatibility

- Support for modern web standards (HTML5, CSS3, JavaScript ES6+).
- Compatible with websites and web applications.
- Proper rendering of multimedia content.

Networking Requirements

10. Internal Networking:

- Virtual network bridge between host and isolated environment.
- Controlled network access for browser (outbound HTTP/HTTPS only).
- Localhost VNC connection for display streaming.
- Blocked direct access to host filesystem and services.

Chapter 04: Feasibility Study

Economical Feasibility

This Project will entirely open-source technologies including

- Firecracker/Cloud Hypervisor/Podman (Apache 2.0 license)
- Chromium browser (BSD license)

No licensing fees or proprietary software costs. Development requires only time investment using freely available tools.

Organizations deploying the solution incur minimal costs:

- Hardware: Standard enterprise workstations (no specialized equipment).
- Infrastructure: No cloud servers required (unlike RBI).
- Licensing: Zero cost (open-source stack).
- Support: Standard IT support staff (no specialized training).

Preventing a single data breach (average cost \$4.88 million) through improved browser security justifies deployment costs many times over. Organizations eliminate recurring RBI subscription fees while achieving equivalent or superior security.

Technical Feasibility

Project requires including:

- Linux system administration and virtualization knowledge.
- Python/C++ programming for system integration.
- Browser technology understanding (Chromium architecture).
- Security and malware analysis fundamentals.

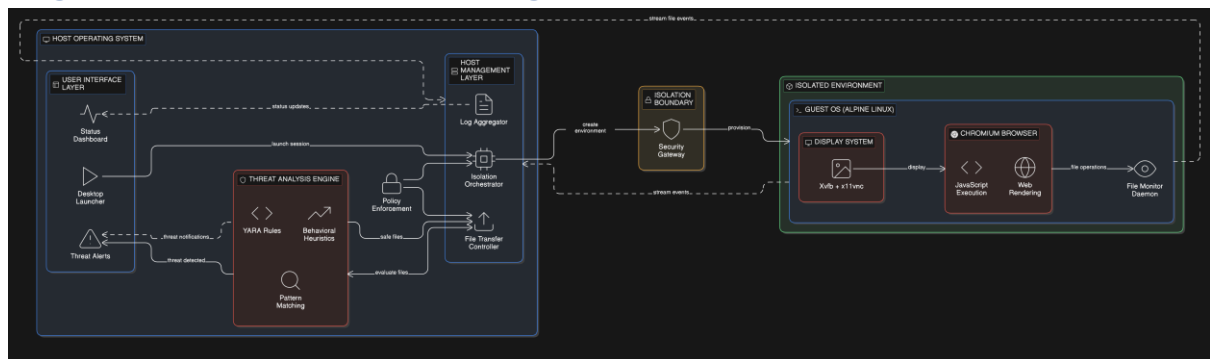
Developer has background in systems programming and security, making implementation technically feasible within project timeline.

All required technologies are mature, well-documented, and actively maintained:

- Firecracker: Production-ready (powers AWS Lambda/Fargate).
- Cloud Hypervisor: Stable (default VMM for Kata Containers).
- Podman: Mature (Red Hat supported, widely deployed).
- Chromium: Industry-standard browser engine.

Chapter 05: System Architecture

High-Level Architectural Diagram



Virtualization Technology Comparison

The system architecture supports three virtualization approaches, each with different trade-offs,

Firecracker MicroVMs:

- Strongest isolation (hardware virtualization, separate kernel)
- Fast boot time (125ms)
- No native display support (requires Xvfb + VNC)
- Linux host only
- Optimal for maximum security

Cloud Hypervisor:

- Hardware virtualization with separate kernel
- Similar performance to Firecracker (100-150ms boot)
- GPU/display support via virtio-gpu
- Linux host only
- Balance of security and functionality

Podman Containers:

- Enhanced container security (rootless, SELinux)
- Instant startup
- Simple X11 display sharing
- Works on Windows and Linux
- Faster development but kernel-level isolation trade-off

Final implementation will select based on deployment requirements, with Podman offering fastest path to working prototype while microVM approaches provide strongest security guarantees.

Chapter 06: Development Tools and Technologies

Programming Languages and Tools

Primary Languages

- **Python 3.10+:** System orchestration, process management, logging infrastructure, and threat analysis engine. Chosen for rapid development, extensive libraries for system interaction, and strong security analysis tools.
- **Bash/Shell Scripting:** Automation scripts for environment setup, VM lifecycle management, and deployment tasks.
- **C/C++:** Performance-critical components such as file system monitoring hooks and low-level system integration. Required for kernel-level monitoring capabilities.

Third-Party Components and Libraries

Virtualization Platforms (Choose One)

- **Firecracker:** Lightweight microVM technology from AWS, providing hardware-level isolation with 125ms boot times
- **Cloud Hypervisor:** Rust-based microVM with GPU support and broader device compatibility
- **Podman:** Enhanced container platform with rootless operation and SELinux integration

Browser Engine

- **Chromium:** Open-source browser engine providing modern web standards support (HTML5, CSS3, JavaScript ES6+)
- Alternative: Firefox/Gecko engine if compatibility issues arise

Display System

- **Xvfb:** X11 virtual framebuffer for headless browser operation
- **x11vnc:** VNC server for streaming display to host
- **TigerVNC/RealVNC:** VNC client for viewing isolated browser

Monitoring and Analysis

- **inotify/fanotify:** Linux kernel APIs for file system event monitoring
- **YARA:** Pattern matching tool for malware detection with extensive rule repository
- **Python watchdog:** High-level file monitoring library
- **psutil:** System and process monitoring utilities

Logging and Data Management

- **Python logging module:** Structured application logging
- **JSON:** Log data format for machine-readable audit trails
- **SQLite:** Local database for event storage and querying

Chapter 07: Implementation Progress

Development Environment Setup

Environment configuration is in progress with foundational infrastructure being established.

Completed Setup:

- Development workstation configured with Ubuntu 22.04 LTS
- Version control repository initialized on GitHub
- Python 3.11 development environment installed with virtual environment
- Basic project structure created with directories for source code, tests, and documentation
- Required system packages installed (build-essential, python3-dev, git)

In Progress:

- Virtualization platform evaluation and selection between Firecracker, Cloud Hypervisor, and Podman
- Display system testing (Xvfb + x11vnc configuration)
- Chromium browser build and customization
- File monitoring daemon prototype development

Pending:

- Integration testing environment setup
- Malware sample collection from public repositories
- Automated testing infrastructure
- Performance benchmarking tools configuration

Implemented Status Overview

Feature	Status	Completion %
Project Infrastructure	Completed	100%
Virtualization Platform Selection	In Progress	40%
Browser Integration	Planned	0%
File Monitoring System	Prototype	15%
Threat Detection Engine	Design Phase	10%
User Interface	Planned	0%
Documentation	In Progress	30%

Challenges Encountered and Solutions

Virtualization Platform Selection

Issue: Multiple viable virtualization options (Firecracker, Cloud Hypervisor, Podman) each with different trade-offs between security, performance, and complexity.

Analysis:

- Firecracker provides strongest isolation but lacks native display support
- Cloud Hypervisor balances security with GPU/display capabilities
- Podman offers fastest development path but kernel-sharing security trade-off

Solution Approach: Implementing modular architecture with abstraction layer supporting multiple backends. Initial prototype will use Podman for rapid development, with architecture allowing future migration to microVM solutions. This enables proving core concepts quickly while maintaining path to stronger isolation.

Display System Integration

Issue: Headless virtualization (especially Firecracker) requires complex display streaming setup using Xvfb and VNC, adding latency and implementation complexity.

Analysis: VNC introduces 50-100ms display latency and requires additional configuration. Podman's X11 socket sharing is simpler but reduces isolation strength. Cloud Hypervisor's virtio-gpu support offers middle ground.

Current Approach: Testing Podman with X11 socket sharing for initial prototype. Documenting VNC configuration for future microVM migration. Evaluating Cloud Hypervisor as potential best-of-both-worlds solution.

File Monitoring Scope

Issue: Determining optimal monitoring granularity - too much logging creates performance overhead, too little misses threats.

Analysis: Monitoring every file operation generates significant data volume. Need intelligent filtering to track security-relevant events while ignoring benign operations.

Solution Strategy:

Implementing tiered monitoring:

1. Always log: Downloads, executions, system file modifications
2. Conditionally log: Large-scale file operations (potential ransomware)
3. Ignore: Browser cache, temporary files, known-safe operations

Using inotify file system events with filtering rules to reduce overhead while maintaining security visibility.

7.5 Current System Limitations

Honestly Acknowledging Current State

Implementation Completeness: System is in early development phase. Core infrastructure established but integration and testing are pending. This interim report documents design, architecture, and planned approach with initial prototyping underway.

Technical Limitations:

- **Display Integration:** VNC streaming approach adds latency (target: <100ms overhead)
- **Platform Support:** Currently targeting Linux hosts; Windows support via WSL2 requires additional testing
- **Threat Detection:** Initial implementation will use rule-based detection; machine learning enhancements deferred
- **Performance Optimization:** Early prototypes will not meet final performance targets; optimization phase scheduled for later sprints

Scope Reductions from Original Plan

- Advanced behavioral analysis using machine learning models moved to future enhancement
- Enterprise management features (centralized policy, fleet management) outside current scope
- Multi-browser support (Firefox, Edge) deferred; focusing on Chromium initially
- Mobile platform support (Android, iOS) not included in current project

Known Issues to Address

- Session persistence and recovery mechanisms not yet designed
- File transfer security (preventing malicious payload insertion) requires implementation
- VNC connection encryption and authentication needs configuration
- Resource cleanup and leak prevention requires thorough testing

Development Risks

- Virtualization platform final selection pending additional testing
- Integration complexity may require architecture adjustments
- Performance targets may need revision based on real-world testing
- Timeline pressure may necessitate further scope prioritization

Chapter 08: Discussion

Summary of the Report

This interim report presents the design and early development of a local browser isolation system addressing the critical trade-off between security effectiveness and operational practicality in current solutions. The project tackles browser-based threats—which account for 12% of all cyberattacks and are increasing 198% year-over-year—through hardware-level isolation combined with comprehensive file monitoring and behavioral threat detection. The system analysis identified fundamental limitations in existing approaches: Remote Browser Isolation provides strong security but introduces prohibitive costs (\$120,000-\$600,000 annually per 1,000 users) and performance degradation (2-5 second page delays), while container-based solutions share host kernels creating escape vulnerabilities. The proposed architecture addresses these gaps through lightweight virtualization technologies (Firecracker, Cloud Hypervisor, or Podman) combined with intelligent file monitoring and YARA-based threat detection.

What Has Changed from the Proposal

Virtualization Platform Approach

Original proposal focused exclusively on Firecracker microVMs as the isolation mechanism. Research revealed display integration challenges (Firecracker lacks graphics support requiring VNC streaming) and discovered Cloud Hypervisor as an alternative offering similar security with native GPU/display capabilities. Additionally, Podman emerged as a pragmatic option for rapid prototyping despite kernel-sharing limitations.

Performance Targets

Proposal specified browser initialization under 150 milliseconds. Implementation research revealed this target applies only to microVM startup time, not complete system initialization including display configuration and browser launch.

Future Plans / Upcoming Work

Immediate Next Steps (March-April 2026)

1. **Finalize Virtualization Platform Selection:** Complete evaluation testing of Podman, Cloud Hypervisor, and Firecracker. Benchmark performance, test display integration, validate isolation strength. Select primary platform with documented fallback options.
2. **Implement Browser Integration:** Build launcher that creates isolated environment, starts browser, configures display streaming (or X11 socket sharing), and presents browser window to user. Target single-click operation hiding complexity.
3. **Develop File Monitoring System:** Complete daemon implementation tracking file operations within isolated environment. Implement event filtering, structured logging, and communication channel to host analysis engine.
4. **Build Threat Detection Engine:** Integrate YARA rules for known malware signatures. Implement behavioral heuristics for ransomware (rapid encryption patterns), information stealers (credential access), and privilege escalation. Configure policy enforcement with automated blocking of detected threats.
5. **Security Testing:** Collect malware samples from public repositories (VirusTotal, MalwareBazaar, EMBER dataset). Execute controlled detonation in isolated environment. Measure detection rates, false positives, and containment effectiveness. Target 95% detection accuracy.

Post-Project Enhancements

- **Windows Native Support:** Port solution to Windows using Hyper-V containers or Windows Sandbox for native operation without WSL2 requirement
- **Multi-Browser Support:** Extend beyond Chromium to Firefox and Edge engines
- **Machine Learning Integration:** Train behavioral models for unknown threat detection and anomaly identification
- **Enterprise Features:** Centralized policy management, fleet monitoring dashboard, SIEM integration
- **Performance Optimization:** Reduce initialization times, minimize resource overhead, optimize display streaming
- **Mobile Platforms:** Investigate isolation approaches for Android and iOS browsers

Research and Publication

- Document security effectiveness through peer-reviewed testing methodology
- Publish open-source implementation enabling community contributions
- Present findings at security conferences or academic venues
- Contribute to broader discourse on cost-effective endpoint security

The project demonstrates that practical, cost-effective browser isolation with strong security guarantees is achievable using open-source technologies. While challenges remain in optimization and feature completion, the core architecture validates the viability of local isolation as an alternative to expensive cloud-based RBI solutions.

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